

Andrew Couperus

Department of Physics and Astronomy
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Curriculum Vitae

Professional Profile: I am an observational astronomer, undergraduate educator, and science communicator, throughout which I also use the lens of astronomy to engage students and the public with climate change and sustainability topics. My research studies the activity of nearby small stars and their exoplanets, with a focus on understanding how exoplanet host stars behave and evolve over time.

Education

- 2018–2025 **PhD - Astrophysics**, Georgia State University, Atlanta, GA.
Defended June 2025 | Research Adviser: Dr. Todd Henry
Thesis: *The Long-Term Stellar Activity Cycles and Magnetic Predictability of Nearby M Dwarfs*
- 2018–2020 **MS - Physics**, Georgia State University, Atlanta, GA.
Concentration in Astronomy | Research Adviser: Dr. Todd Henry
- 2014–2017 **BS - Physics (with Great Distinction)**, Clarkson University, Potsdam, NY.
Minor in Mathematics | Research Adviser: Dr. Joshua Thomas

Appointments

- August 2026 to present **Associate Director of Outreach and Education**, Limited-Term Faculty, Department of Physics and Astronomy, Georgia State University, Atlanta, GA.
– Executing a multi-pronged position involving: (1) Taking over management and development of the Department's educational [HyperPhysics website](#). (2) Assisting the Department's numerous outreach efforts, including leading a collaboration between the Department, [Pink STEM](#), and Georgia's Museum of Aviation. (3) Teaching introductory physics content, including taking over the Department's *Physics of Music and Speech* course.
– Teaching guest lectures on science communication and climate change topics for seminars.
- ongoing **Research Affiliate**, REsearch Consortium On Nearby Stars (**RECONS** - recons.org).
- 2025–2026 **Visiting Assistant Professor of Astronomy**, Smith College, Northampton, MA.
– Member of the Five College Astronomy Department (FCAD)
- 2018–2025 **Graduate Research Assistant**, Georgia State University.

Professional Experience

Teaching

- 2025–2026 **Visiting Assistant Professor of Astronomy**, Smith College.
– Taught 5 undergrad courses over the 2025–2026 academic year, including (AST 100) *A Survey of the Universe*, (AST 235) *Introduction to Stellar Structure*, and the interdisciplinary (AST 214) *Astronomy & Public Policy*. Also designed, successfully proposed for, and taught a new intermediate-level research-focused course, (AST 221) *Current Problems in Low-Mass Stars*.
– Emphasis on active learning, un-grading, inclusive teaching, and societally-relevant content.
– Consistently received perfect and near-perfect scores on anonymous student evaluations, alongside *many* [qualitative endorsements from students](#).

- 2025 **Workshop Participant**, AAS #245, *Increasing Student Learning and Inclusion in Your Classroom: Strategies from the Faculty Teaching Institute*.
- 2018–2021 **Graduate Teaching Assistant**, Georgia State University.
- Taught 16 lab sections across eight semesters for the undergraduate intro astronomy courses (ASTR 1010) *Astronomy of the Solar System* and (ASTR 1020) *Stellar and Galactic Astronomy*.
 - Helped improve in-person lab activities and contributed to training new TAs.
 - Completed online teaching training, developed new online lab materials, led online groups of new TAs, and helped coordinate transition to online lab teaching during the pandemic.

Research

- 2018–2025 **Graduate Research Assistant**, Georgia State University.
- Investigated nearby low-mass stars, particularly their stellar magnetic activity, long-term activity cycles, rotation, variability, X-ray emission, exoplanets, and multiplicity. Managed six observing campaigns obtaining short- and long-baseline optical photometry, optical spectroscopy, X-ray imaging, speckle imaging, and ground-based astrometry, alongside a large breadth of archival data sources including *Gaia*, *TESS*, *Kepler*, ZTF, ASAS-SN, and 2MASS. Applied Bayesian numerical methods via Gaussian Processes and MCMC analysis methods.
 - Co-advised undergraduate research student, Summer 2022.
- 2016–2017 **Undergraduate Research Assistant**, Clarkson University.
- Helped implement and calibration a new LHIRES III spectrograph at Reynolds Observatory.
 - Completed spectral observations and analysis for ~ 40 nights of data to refine orbital measurements of high-mass stellar binaries.

Observing

- 2026–now **Chandra X-ray Observatory**, near-Earth orbit.
- 377 ksec – PI of accepted Cycle 28 General Observer (GO) proposal ID 28200109, pending observation.
- *Observing time award monetary equivalent*: $\sim \$800,000$
- 2025A **Canada-France-Hawai'i Telescope 3.6m**, Maunakea, Hawai'i.
- 3.6 hrs – Awarded snapshot time with the SPIRou spectropolarimetry instrument as science Co-I.
- *Observing time award monetary equivalent*: $\sim \$8,000$
- 2019–2025 **RECONS CTIO/SMARTS 0.9m Program Support**, La Serena, Chile.
- Regularly assisted photometric and astrometric observations and analysis for the RECONS multi-decade 0.9m program on nearby stars.
 - Coordinated simultaneous 0.9m and 1.5m observations for a targeted multi-messenger study.
- 2019–2023 **CTIO/SMARTS 0.9m**, La Serena, Chile.
- 68 nights – Experience carrying out multiple 12–20 night in-person observing runs.
- 36 nights awarded competitively from NOIRLab proposal 2023A-549259 as PI, with another 36 nights awarded as PI via proposals over 2020–2021 but lost due to the COVID-19 pandemic.
 - *Observing time award monetary equivalent*: $\sim \$50,000$
- 2019–2023 **CTIO/SMARTS 1.5m**, La Serena, Chile.
- 203 hrs – Observations with the high-resolution CHIRON spectrograph through RECONS/GSU time.
- 2021–2022 **XMM-Newton**, near-Earth orbit.
- 13 ksec – Awarded low-priority time from GO proposal ID 088170 as Co-I.
- 2020–2022 **Chandra X-ray Observatory**, near-Earth orbit.
- 188 ksec – Awarded as science lead Co-I via GO proposal ID 22200260, alongside a \$65,845 SAO grant.
- *Observing time award monetary equivalent*: $\sim \$400,000$
- 2019 **Apache Point Observatory - ARC 3.5m**, Sunspot, NM.
- 3 half-nights – Trained with the high-resolution ARCES spectrograph.

- 2019 **Hard Labor Creek Observatory - Miller 0.61m**, Rutledge, GA.
 3 nights – Collected photometric observations of a rotating asteroid.
- 2016–2017 **Reynolds Observatory - 12in Meade**, Potsdam, NY.
 ~20 nights – Acquired optical spectra for a multi-institution project including citizen scientists.

Industry

- 2017–2018 **Customer Service Technician**, Frazer Computing, Canton, NY.
 – Worked in a team-based technical environment to support software and characterize user bugs.

Publications

3 first-author (2 published, 1 fully drafted), 8 co-authored

- Fully drafted & in prep **Andrew A. Couperus**, Todd J. Henry, Eliot H. Vrijmoet, et al., *The Solar Neighborhood. LVI. New Photometric Stellar Activity Cycles in Fully Convective M Dwarfs Reveal Cycle Periods Beyond Two Decades and an Occurrence Rate Above 5%*, drafted and in prep.
- Submitted Madison R. LeBlanc, Tim M. Johns, Todd J. Henry, ... [including **Andrew A. Couperus**], et al., *The Solar Neighborhood LVI: The RMSTAR Catalog of the Nearest 3352 M Dwarf Systems and 305 of their Wide Companions*, submitted to AJ.
- 2026 **Andrew A. Couperus**, Todd J. Henry, Aman Kar, et al., *The Solar Neighborhood. LIII. M Dwarf Twin Binaries — One in Five Twin Sibling Pairs Are Mismatched in Activity and/or Rotation*, AJ, 171, 168, available at [ADS](#) or [AJ](#), 45 pages.
- 2025 **Andrew A. Couperus**, Todd J. Henry, Rachel A. Osten, et al., *The Solar Neighborhood. LII. M Dwarf Twin Binaries — Presumed Identical Twins Appear Fraternal in Variability, Rotation, $H\alpha$, and X-rays*, AJ, 169, 41, available at [ADS](#) or [AJ](#), 34 pages.
- 2024 T.A. Rector, L. Barbier, **Andrew A. Couperus**, et al., *Climate Change Task Force Report for the American Astronomical Society*, arXiv, [arXiv:2406.10451](#).
 – Aided in AAS emissions assessment, membership climate survey, and writing of report.
- 2024 Aman Kar, Todd J. Henry, **Andrew A. Couperus**, et al., *The Solar Neighborhood LI: A Variability Survey of Nearby M Dwarfs with Planets from Months to Decades with TESS and the CTIO/SMARTS 0.9 m Telescope*, AJ, 167, 196, [doi:10.3847/1538-3881/ad2ddc](#).
 – Aided development and guidance of project, some analysis codes, and writing of paper.
- 2022 Wei-Chun Jao, **Andrew A. Couperus**, Eliot H. Vrijmoet, et al., *Estimating the Convective Turnover Time*, ApJ, 940, 145, [doi:10.3847/1538-4357/ac9cd8](#).
 – Aided discussions of project, interpretation of analysis, and writing of paper.
- 2021 Joshua D. Thomas, Noel D. Richardson, J. J. Eldridge, ... [including **Andrew A. Couperus**], et al., *The orbit and stellar masses of the archetype colliding-wind binary WR 140*, MNRAS, 504, 5221, [doi:10.1093/mnras/stab1181](#).
 – Acquired many observations and processed a portion of the spectra for RV analyses.
- 2020 Douglas R. Gies, Kathryn V. Lester, Luqian Wang, **Andrew A. Couperus**, et al., *Spectroscopic Detection of the Pre-White Dwarf Companion of Regulus*, ApJ, 902, 25, [doi:10.3847/1538-4357/abb372](#).
 – Aided preliminary RV analyses of the system.

- 2020 Emily A. Gilbert, Thomas Barclay, Joshua E. Schlieder, ... [including **Andrew A. Couperus**], et al., *The First Habitable-zone Earth-sized Planet from TESS. I. Validation of the TOI-700 System*, AJ, 160, 116, [doi:10.3847/1538-3881/aba4b2](https://doi.org/10.3847/1538-3881/aba4b2).
– Acquired absolute photometric observations to help validate the host star properties.
- 2018 Rachel A. Johnson, Noel D. Richardson, Anthony F. J. Moffat, ... [including **Andrew A. Couperus**], et al., *An Updated Ephemeris for the Single-lined Orbit of the Supergiant μ Sagittarii*, RNAAS, 2, 138, [doi:10.3847/2515-5172/aad6ed](https://doi.org/10.3847/2515-5172/aad6ed).
– Acquired many observations and processed a portion of the spectra for RV analyses.

Awards & Funding (Total: \$70,905)

- 2025 **\$5,060**, NASA SCoPE Seed Grant Proposal (awarded but declined due to scheduling).
Engaging Children and Adults with (Exo)Planetary Astronomy at Local Libraries
- 2021–2024 **\$65,845**, Smithsonian Astrophysical Observatory, Co-I, via Chandra prop. ID 22200260.
Fraternal or Identical? The Magnetic Properties of M Dwarf Twins
- 2025 **Outstanding Advanced Astronomy Graduate Student Award**, GSU.
- 2021 **Outstanding Junior Astronomy Graduate Student Award**, GSU.
- 2020 **Exceptional Department Service Award**, GSU.
- 2020 **Outstanding Astronomy Graduate Teaching Assistant Award**, GSU.
- 2020 **Honorable Mention**, NSF Graduate Research Fellowship Program.
- 2015–2017 **Presidential Scholar**, Clarkson University.
- 2014 **College Academic Award in Engineering Individual Studies**, SUNY Canton.

Professional Service

- 2020–2025 **Graduate Student Mentor**, AstroPALs, GSU.
– Directly mentored 2 students, developed and led 5 focus group sessions and co-led several others, and regularly aided the Astronomy Peer Advising Leaders (AstroPALs) steering committee.
- 2023 **Astronomy Student Representative**, Department Graduate Committee, GSU.
- 2018–2022 **Stellar Journal Club Rotating Discussion Leader**, GSU.
- 2020 **Astro/Physics Graduate Student DEI Committee Member**, GSU.

(See *Climate Change Education, Action, & Service* for additional service items.)

Climate Change Education, Action, & Service

- 2025–present **Chair, Sustainability Committee**, American Astronomical Society (AAS).
– Leading a team of ~10 astronomers in various projects related to sustainability in astronomy.
– Routinely lead talks, events, recruitment efforts, and operational logistics for the Committee.
– PI of successful proposal for and operational lead of the Special Session *What Astronomers Can Do About Climate Change: Infrastructure, Education, and Communication* at AAS #248.
- 2024–2025 **Member, Sustainability Committee**, AAS.
- 2021–present **Member**, Astronomers for Planet Earth (A4E), and A4E Education Working Group.
– *Engaging Astronomy Students with a World Climate Simulation* (article available [here](#))

2022–2025 **Astronomy × Climate Change Guest Lecturer**, GSU.

- Taught guest lectures for 3 graduate and undergraduate astronomy classes to discuss content at the intersection of astronomy and climate change.
- Provided help for ~10 others to include such content in their classes and research efforts.

2024 **Invited Speaker**, *Climate Change and the American Astronomical Society*, GSU.

2024 **Workshop Participant**, AAS #243, *Saving Astronomy Workshop: Light Pollution, Satellite Constellations, and Climate Change*.

- Worked with interdisciplinary professionals spanning architects to rocket scientists in order to develop action items for advancing light pollution and satellite contamination mitigation efforts.

2022–2024 **Climate Change Task Force Member**, AAS.

- [report](#) – Helped assess AAS CO2 emissions, survey AAS membership regarding climate action, investigate virtual meeting methods, and write report with recommendations for AAS leadership.

2021 **Completed Climate Leadership Training**, The Climate Reality Project.

Public Outreach

[flyer](#) 2026 **Volunteer Presenter**, Astrophysics Speaker Series, EcoTarium Museum, Worcester MA.

2025–2026 **Open Night Assistant**, McConnell Rooftop Observatory, Smith College.

[article](#) 2025 **S&T Magazine Focal Point Author**, November 2025 Issue, *A Flare in Real Time*.

2024–2025 **Volunteer Presenter** ×2, Three Taverns Brewery: Astronomy Night Lecture Series.

2018–2025 **Open Night Assistant**, Hard Labor Creek Observatory, GSU.

2024 **Science Activity Leader**, John Lewis Elementary School STEM Night.

2021 & 2022 **GSU Committee Member and Activity Leader**, Atlanta Science Festival.

[art](#) 2021 **Science Partner**, Science.Art.Wonder, Georgia Institute of Technology.
– Collaborated with a digital artist to convey astronomy concepts.

2019 **Program Assistant**, Georgia Science Olympiad Regional Tournament, GSU.

2019 **Science Activity Leader**, Trip Elementary School Science Night.

2017 **Color Images of the Orion Nebula**, Reynolds Observatory, Clarkson University.
– Created new composite color images of the Orion Nebula for use in public engagement.

2016–2017 **Open Night Assistant**, Reynolds Observatory, Clarkson University.

Summer 2016 **Mentor & Program Aid**, IMPETUS High School Program, Clarkson University.

Technical Skills

Proficient Python, Jupyter Notebooks, IRAF, LaTeX, Windows, Linux

Introductory IDL, C++, MATLAB, Bash scripting, HTML

2012 Certified Microsoft Office Specialist in Word, PowerPoint, and Excel.

Presentations 16 talks (3 invited), 6 posters

Talks

2026 *What Astronomers Can Do About Climate Change: Education and Communication*.
| AAS Meeting #248

(invited) 2025 *The Magnetic Predictability and Long-Term Stellar Activity Cycles of Nearby M Dwarfs*.
| Five College Astronomy Department Colloquium

- 2025 *The Interconnection of Astronomy and Climate Change.*
| GSU Graduate Conference for Research, Scholarship, and Creative Activity #3
- 2025 *The AAS Climate Change Task Force Report.*
| AAS Meeting #245
- 2025 *The Magnetic Predictability and Stellar Activity Cycles of Nearby M Dwarfs.*
| AAS Meeting #245
- 2024 *Twin M Dwarfs Appear Both Fraternal and Identical in Activity and Rotation.*
| GSU Stellar Symposium
- (invited) 2024 *Climate Change and the American Astronomical Society.*
| GSU Department Seminar
- abstract 2024 *Seeing Double: Are Twin M Dwarfs Fraternal or Identical in Activity and Rotation.*
| AAS Meeting #243, 254.05
- 2023 *Seeing Double: Are Twin M Dwarfs Fraternal or Identical in Activity.*
| GSU Stellar Symposium
- (invited) 2022 *Twinkle Twinkle Little Star ET Wonders How You Are.*
| STScI Special Seminar
- 2022 *M Dwarf Stellar Activity — A Coming-of-Age Story.*
| Clarkson University Summer Undergraduate Research Program
- 2022 *M Dwarf Stellar Activity — A Coming-of-Age Story.*
| GSU Galaxies to Gluons Summer Seminar Series
- abstract 2022 *Stellar Cycles in Fully Convective M Dwarfs: Astronomy Beyond a Funding Cycle.*
| Skumanich Conference, id.29
- 2021 *Twinkle Twinkle Little Star ET Wonders How You Are.*
| GSU Undergraduate Research Program Summer Seminar Series
- abstract 2020 *Characterizing M Dwarf Stellar Cycles with Two Decades of RECONS Data.*
| AAS Meeting #236, 319.01
- 2016 *Benchmarking of the Shelyak LHIRES III Spectrograph.*
| Clarkson SURE Conference

Posters

- abstract 2026 *Four Student-Led Projects Reveal New Insights into M Dwarf Stellar Activity Cycles and Their Impact on Exoplanet Habitability and Detection.*
| AAS Meeting #248, 400.13
- poster 2024 *Twin M Dwarfs Appear Both Fraternal and Identical in Activity and Rotation.*
| Cool Stars 22 Conference
- abstract 2022 *The Long-Term Photometric Variability of Nearby M Dwarfs and Exoplanet Hosts.*
| AbSciCon2022 Conference
- poster 2021 *Twinkle Twinkle Little Star: ET Wonders How You Are.*
| Cool Stars 20.5 Conference
- abstract 2021 *Twinkle Twinkle Little Star: ET Wonders How You Are.*
| AAS Meeting #237, 141.04
- 2016 *The Science at Clarkson's Reynolds Observatory.*
| Astronomical Society of New York Conference

Professional References

1. Dr. Todd Henry, RECONS & Georgia State University, thenry88@gsu.edu
2. Dr. James Lowenthal, Smith College, jlowenth@smith.edu
3. Dr. Rachel Osten, STScl & Johns Hopkins University, osten@stsci.edu
4. Dr. Travis Rector, University of Alaska Anchorage, tarector@alaska.edu
5. Dr. Wei-Chun Jao, RECONS & Georgia State University, wjao@gsu.edu